

Rishav Sood

6/9/26

Transcriptomic Analysis of GPR15L in DSS-Induced Colitis Using Public RNA-Seq Data

Introduction:

Inflammatory bowel disease (IBD) is a long-term inflammatory disorder of the gastrointestinal tract. It involves out-of-control immune responses, problems with the epithelial barrier, and changes in how the body interacts with intestinal bacteria. While the exact ways IBD works are not fully understood, many studies show that both innate and adaptive immune pathways play a role in disease progression.

G protein-coupled receptor 15 ligand (GPR15L) is a protein that helps control immune cell movement in the intestinal lining. Previous research indicates that GPR15 and GPR15L help keep intestinal immune balance and may affect inflammatory responses during colitis. The Gene Expression Omnibus (GEO) study analyzed in this project looked at the role of GPR15L in dextran sodium sulfate (DSS)-induced colitis using wild-type and GPR15L knockout mice. The study also examined the connection between GPR15L, the composition of intestinal bacteria, and disease severity.

RNA sequencing (RNA-seq) offers a robust way to study changes in gene expression linked to disease and genetic alterations. By identifying genes that are expressed differently and biological pathways that are enriched, transcriptomic analysis can uncover the molecular processes involved in disease development. In this project, we examined publicly available RNA-seq data from GEO using R and DESeq2. Our goal was to investigate how the deletion of GPR15L affects gene expression patterns in colon tissue and lamina propria mononuclear cells (LPMCs) during DSS-induced colitis.

Hypothesis:

Deletion of GPR15L changes the transcriptional programs related to intestinal inflammation, immune regulation, and host-microbiota interactions during DSS-induced colitis. Specifically, lacking GPR15L was expected to impact genes linked to immune signaling, maintaining the epithelial barrier, and adaptive immune responses.

Procedure:

RNA-seq count data were obtained from the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO). The dataset included expression profiling through high-throughput sequencing from *Mus musculus* samples collected during DSS-induced colitis experiments. The samples collected were organized in four experimental groups: Colon WT DSS, Colon GPR15L KO DSS, LPMC WT DSS, and LPMC GPR15L KO DSS. Each experimental group has three biological replicates. Genes that had an adjusted p-value < 0.05 and an absolute log₂ fold change > 1 were considered significantly differentially expressed.

Volcano plots were created with ggplot2 to show differential gene expression. Log₂ fold change appeared on the x-axis, while negative log₁₀ adjusted p-value was on the y-axis. Significant genes were marked to reveal transcripts with both large expression changes and strong statistical significance. Variance-stabilized counts were created with DESeq2. The pheatmap feature was used to make heatmaps that show expression patterns among the top differentially expressed genes. Hierarchical clustering helped us find relationships among biological replicates and experimental groups.

Significant Ensembl gene identifiers were converted to gene symbols with the org.Mm.eg.db annotation database. Gene functions were explored using literature and public genomic databases. Gene Ontology (GO) Biological Process enrichment and KEGG pathway enrichment analyses were done with clusterProfiler. Significant genes were changed to Entrez identifiers before the enrichment analysis. Adjusted p-values helped identify enriched pathways, which were shown with dot plots.

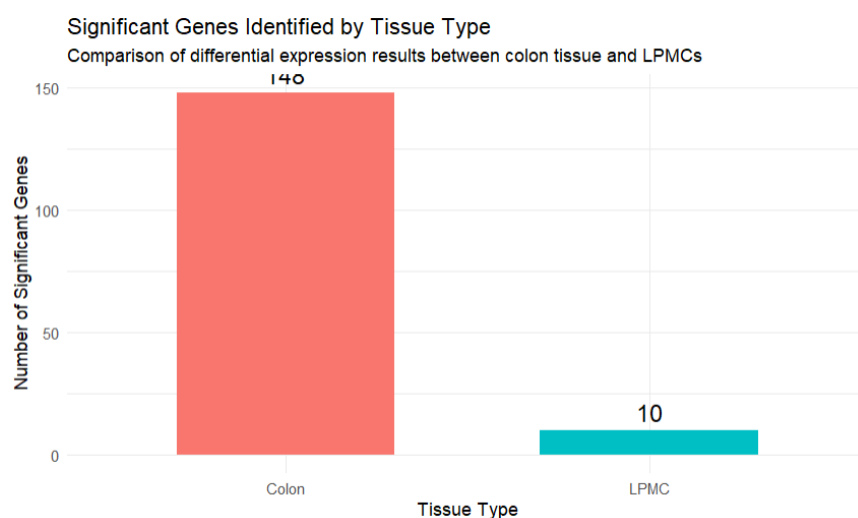
Results:

Figure 1. *Number of Differentially Expressed Genes Identified in Colon Tissue and LPMC.* Differential expression analysis identified 148 significant genes in colon tissue and 10 significant genes in lamina propria mononuclear cells (LPMCs) using thresholds of adjusted p-value < 0.05 and $|\log_2 \text{fold change}| > 1$. These findings indicate that GPR15L deletion has a substantially greater impact on transcriptional programs in colon tissue than in LPMCs during DSS-induced colitis.

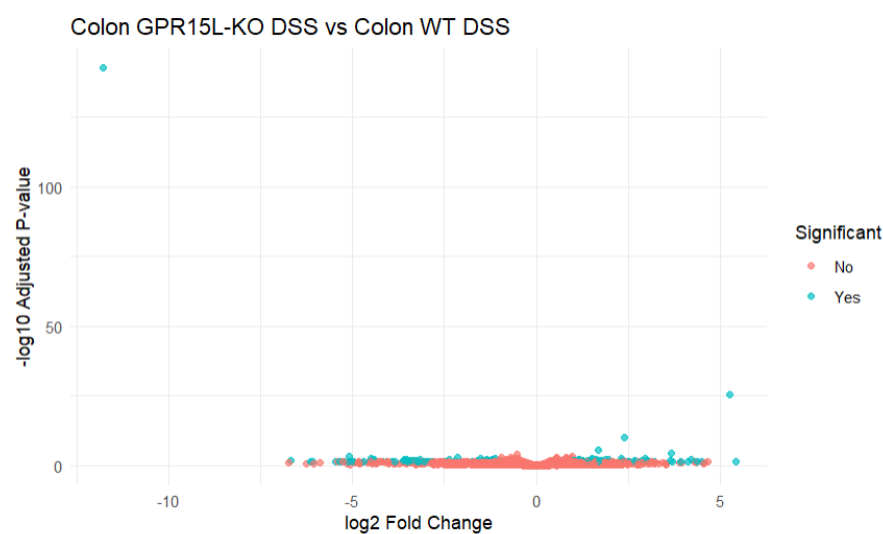


Figure 2. *Differential Gene Expression in Colon Tissue Following GPR15L Deletion.* Volcano plot showing differential gene expression between Colon GPR15L-KO DSS and Colon WT DSS mice. Significant genes ($\text{padj} < 0.05$ and $|\log_2\text{FC}| > 1$) are highlighted. *Gpr15l* was the most significantly downregulated gene, confirming successful deletion of the target gene.

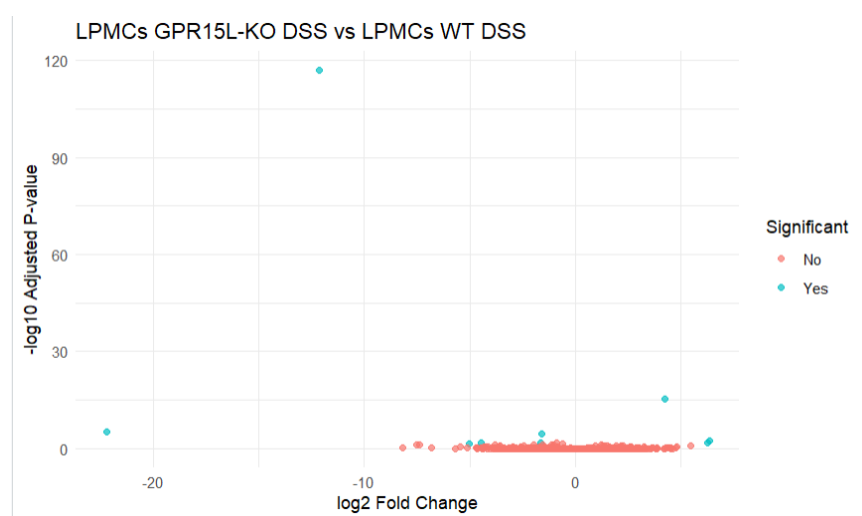


Figure 3. *Differential Gene Expression in LPMC Tissue Following GPR15L Deletion.* Volcano plot showing differential gene expression between LPMC GPR15L-KO DSS and LPMC WT DSS mice. Significant genes ($\text{padj} < 0.05$ and $|\log_2\text{FC}| > 1$) are highlighted. *Gpr15lg* was the most significantly downregulated gene, confirming successful deletion of the target gene.

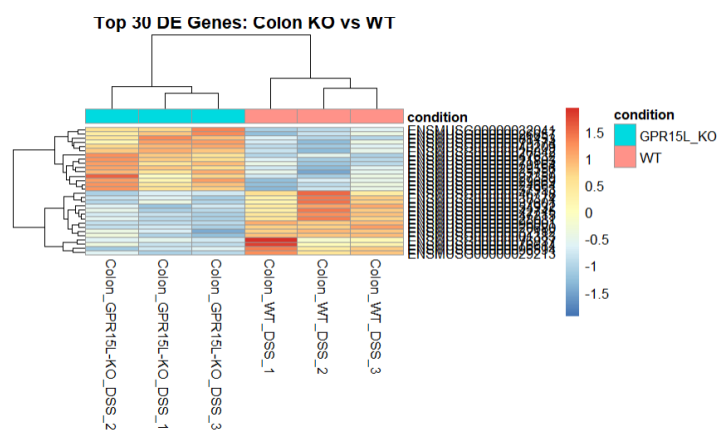


Figure 4. *Clustering of Differentially Expressed Genes in Colon Tissue.* Heatmap of the top 30 differentially expressed genes in colon tissue. Hierarchical clustering separated WT and GPR15L-KO samples into distinct groups, indicating consistent transcriptional differences between genotypes.

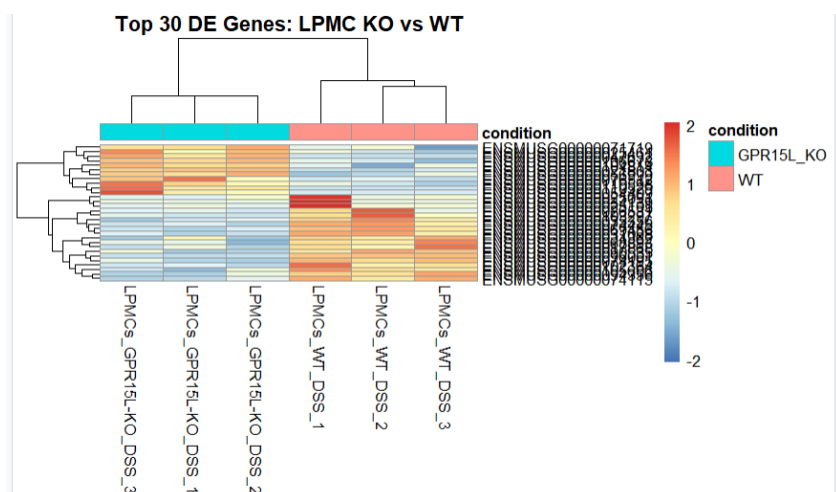


Figure 5. *Clustering of Differentially Expressed Genes in LPMC Tissue.* Heatmap of the top 30 differentially expressed genes in LPMC tissue. Hierarchical clustering separated WT and GPR15L-KO samples into distinct groups, indicating consistent transcriptional differences between genotypes.

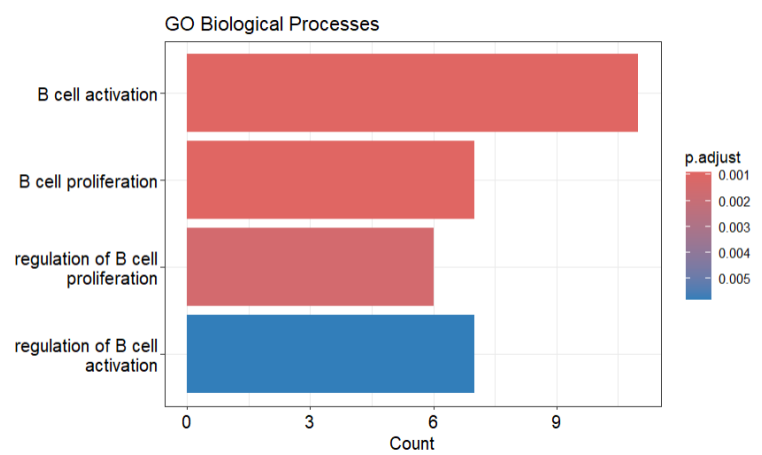


Figure 6. *Gene Ontology Enrichment Analysis of Colon Differentially Expressed Genes*. GO Biological Process enrichment analysis identified significant enrichment of pathways related to B-cell activation, B-cell proliferation, regulation of B-cell activation, and B-cell receptor signaling, indicating altered adaptive immune responses in GPR15L-deficient mice.

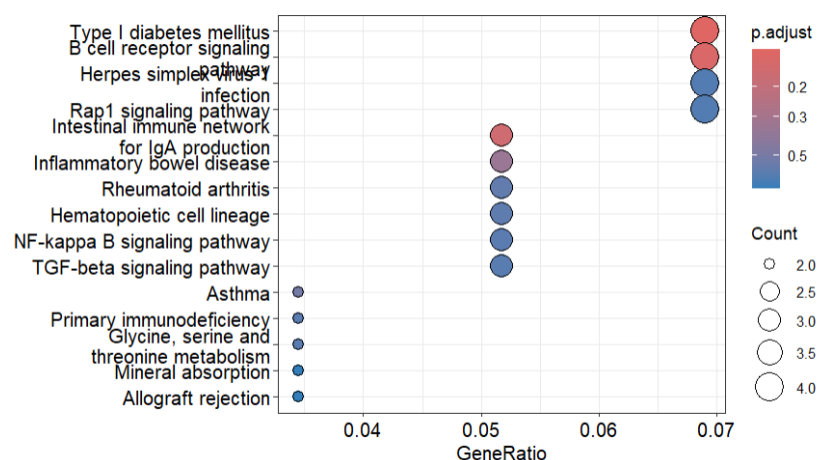


Figure 7. *KEGG Pathway Enrichment Analysis*. KEGG pathway enrichment analysis revealed significant enrichment of immune-related pathways, including inflammatory bowel disease (IBD), intestinal immune network for IgA production, B-cell receptor signaling, NF- κ B signaling, TGF- β signaling, and hematopoietic cell lineage pathways. These results suggest that GPR15L influences adaptive immune responses, mucosal immunity, and inflammatory signaling during DSS-induced colitis. The enrichment of IBD-associated pathways further supports a role for GPR15L in the pathogenesis of intestinal inflammatory disease.

Table 1. Top Differentially Expressed Genes in Colon Tissue

Gene	log2FC	padj	Biological Function
Gpr15lg	-11.81	1.66×10^{-143}	GPR15 ligand; regulator of intestinal immune homeostasis
Cdhr1	5.27	2.59×10^{-26}	Cell adhesion and epithelial organization
Fgfbp3	2.41	1.12×10^{-10}	Growth factor signaling and tissue repair
Tlr5	1.70	2.74×10^{-6}	Recognition of bacterial flagellin and innate immunity
Igkv8-19	-5.10	3.99×10^{-4}	Immunoglobulin variable region; B-cell responses
Afp	-2.15	1.07×10^{-3}	Developmental and metabolic regulation
Slc27a2	2.96	2.60×10^{-3}	Lipid metabolism and epithelial physiology
Bst2	-1.51	2.60×10^{-3}	Interferon signaling and innate immunity
Nme1	-1.11	2.60×10^{-3}	Cell signaling and proliferation
Iglc2	-4.51	2.60×10^{-3}	Immunoglobulin light chain; adaptive immunity
A4gnt	2.32	2.77×10^{-3}	Glycosylation and mucosal protection
Tspan12	1.52	4.82×10^{-3}	Cell membrane organization and signaling
Id1	-1.19	5.49×10^{-3}	Transcriptional regulation and cell differentiation
Ambp	4.21	7.56×10^{-3}	Protease inhibition and tissue protection

P2rx1	1.88	7.56×10^{-3}	Purinergic signaling and inflammation
Hif3a	1.63	7.58×10^{-3}	Hypoxia response and cellular adaptation

* Positive $\log_2$ fold-change values indicate higher expression in GPR15L knockout mice, while negative values indicate lower expression relative to wild-type controls

Table 2. Top Differentially Expressed Genes in LPMC Tissue

Gene	log2FC	padj	Biological Function
Gpr15l	-12.14	9.00×10^{-118}	Immune cell trafficking and intestinal immune regulation
Cdhr1	4.22	7.01×10^{-16}	Cell adhesion and epithelial organization
Serpini2	-22.20	8.82×10^{-6}	Protease inhibition and tissue protection
C1qa4b	-1.59	2.68×10^{-5}	Complement activation and innate immunity
Igkv9-123	6.32	5.21×10^{-3}	B-cell receptor diversity and antibody production
Igkv8-28	-4.47	2.22×10^{-2}	Adaptive immune responses
Unknown	-1.65	2.22×10^{-2}	Unknown
Rdh16	-0.91	2.22×10^{-2}	Retinoid metabolism and mucosal immune homeostasis
Unknown	6.26	2.22×10^{-2}	Unknown

Igkv2-109	-5.04	4.94×10^{-2}	Adaptive immunity and antibody generation
-----------	-------	-----------------------	---

* Positive $\log_2$ fold-change values indicate higher expression in GPR15L knockout mice, while negative values indicate lower expression relative to wild-type controls

Discussion:

The results support the idea that GPR15L affects gene programs related to intestinal inflammation and immune control. The significant decrease of Gpr15l in knockout mice shows that the genetic deletion worked and affirms the analysis method. Colon tissue showed many more changes in gene expression than LPMCs, indicating that GPR15L has a bigger effect on local intestinal tissue than on overall immune-cell gene programs.

Several differentially expressed genes suggest changes in epithelial barrier function and interactions with microbiota. Increased Tlr5 expression indicates a better recognition of bacterial products. Higher levels of Cdhr1 and Fgfbp3 expression point to possible shifts in how the epithelium repairs itself and remodels tissue. Pathway enrichment analysis found that B-cell activation and B-cell receptor signaling are the most affected biological processes. These findings imply that GPR15L plays a role in controlling the adaptive immune response during DSS-induced colitis. This idea is also backed by the presence of several immunoglobulin variable-region genes among the significant LPMC transcripts.

KEGG pathway enrichment further demonstrated enrichment of pathways directly related to inflammatory bowel disease, NF- κ B signaling, TGF- β signaling, and intestinal immune network for IgA production. These pathways are central regulators of intestinal inflammation, immune tolerance, and host-microbiota interactions. The enrichment of these pathways suggests that GPR15L influences multiple aspects of intestinal immune homeostasis beyond B-cell activation alone.

The lack of significant enrichment in LPMCs likely comes from the relatively small number of changed genes, not from a lack of importance. Overall, these findings suggest that GPR15L acts as a key regulator of intestinal immune balance and may affect interactions between adaptive immunity, epithelial barrier maintenance, and microbial sensing systems. Future studies could include more GEO datasets, conduct comparisons

across different studies, explore specific expression patterns in different cell types, and create interactive dashboards to help visualize transcriptomic changes in gene expression.

Several limitations should be considered. This study utilized publicly available RNA-seq data and relied on computational analyses rather than experimental validation. In addition, only ten significant genes were identified in LPMCs, limiting the statistical power of pathway enrichment analyses in this tissue. Future studies incorporating additional biological replicates, validation experiments, and integration of multiple transcriptomic datasets would strengthen these findings.

Conclusion:

Transcriptomic analysis of public GEO RNA-seq data showed that deleting GPR15L significantly changes gene expression during DSS-induced colitis. Colon tissue underwent extensive transcriptional remodeling, with 148 significant genes identified and pathways related to B-cell activation and adaptive immunity becoming more prominent. Heatmap and clustering analyses revealed a clear separation between wild-type and knockout samples, showing consistent biological differences. The differentially expressed genes pointed to immune regulation, epithelial repair, microbial recognition, and host-microbiota interactions as key processes impacted by GPR15L deficiency. These findings highlight GPR15L's role in regulating intestinal immune balance and suggest that altered GPR15L signaling may contribute to the development of inflammatory bowel disease.

References:

1. and, S. (2026). *GeneCards - The Human Gene Database*. Genecards.org.
<https://www.genecards.org>
2. *GEO Accession viewer*. (2024). Nih.gov.
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE327543>
3. *MGI-Mouse Genome Informatics-The international database resource for the laboratory mouse*. (n.d.). [Www.informatics.jax.org](http://www.informatics.jax.org). <https://www.informatics.jax.org>
4. National Center for Biotechnology Information. (2019). *Home - Gene - NCBI*. Nih.gov.
<https://www.ncbi.nlm.nih.gov/gene/>
5. UniProt. (2019). *UniProt*. Uniprot.org. <https://www.uniprot.org>